EXP NO:18 Experiment to understand KNN algorithm for a given dataset

AIM:

To make an Experiment to understand KNN algorithm for a given dataset

PROGRAM:

```
from google.colab import files
uploaded = files.upload()
import numpy as np
import pandas as pd
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
df.columns = df.columns.str.strip()
df.info()
print(df.iloc[:, -1].value_counts())
print(df.head())
features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=0.2, random_state=0
)
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)
print("Training Accuracy:", model_KNN.score(xtrain, ytrain))
print("Testing Accuracy:", model_KNN.score(xtest, ytest))
from sklearn.metrics import confusion_matrix
print("Confusion Matrix:")
print(confusion_matrix(label, model_KNN.predict(features)))
from sklzarn.metrics import classification_report

print("Classification Report:")

print(classification_report(label, model_KNN.predict(features)))
```

OUTPUT:

```
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
species
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
Training Accuracy: 0.95
Testing Accuracy: 0.9666666666666667
Confusion Matrix:
[[50  0  0]
 [ 0 45  5]
 [ 0  2 48]]
Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        50
  versicolor       0.96      0.90      0.93        50
   virginica       0.91      0.96      0.93        50

    accuracy                           0.95       150
   macro avg       0.95      0.95      0.95       150
weighted avg       0.95      0.95      0.95       150
```

RESULT:

Thus the experiment to understand KNN algorithm for a given data

Set is done.